

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

***CO3:** Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour
Total Marks: 30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I Dharshan B ¹⁹²³¹²¹⁵⁰ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Dharshan

*18 + 18 + 22 + 18 + 13
= 89*

14

CO3 AT3 - Resolution Algorithm Implementation

1) Library Late Fine

Let: L = Book is returned late
 F = fine

1. propositional logic:-

- * If book is returned late, then fine is charged: $L \rightarrow F$
- * Book was returned late: L
- * Goal: F

2. CNF and Negation of Goal:-

- * $L \rightarrow F \equiv \neg L \vee F$
- * $L \equiv L$
- * Negated goal: $\neg F$

3. Resolution:-

Resolve C1 and C2: $(\neg L \vee F), L \rightarrow F$

new clause: F

Now resolve F with $\neg F$: $F, \neg F \rightarrow \square$

4. Empty Clause: $\square$

2) Flight Delay Notification

Let: D = Flight 202 is delayed, N = passengers are notified

1. propositional logic:-

- * $D \rightarrow N$
- * Goal: N

2. CNF and Negation

* $D \rightarrow N \equiv \neg D \vee N$

* $D \equiv D$

* Negated goal = $\neg N$

3. Resolution:

Resolve C_1 and $C_2: (\neg D \vee N), D \rightarrow N$

Now resolve N with $\neg N: N, \neg N \rightarrow \square$

4. Empty clause: $\square$

Conclusion:-

Therefore, passengers are notified

3) Employee certification.

Let: T = Kevin completed Training

C = Kevin receives certification

Logic:-

* $T \rightarrow C$

* Goal: C

CNF and Negation:-

* $T \rightarrow C \equiv \neg T \vee C$

* $T \equiv T$

Negated goal = $\neg C$

clauses:-

* $C_1: \neg T \vee C$

* $C_2: T$

* $C_3: \neg C$

Resolution:-

Resolve C_1 & $C_2: (\neg T \vee C), T \rightarrow C$

Then resolve C with $\neg C: C, \neg C \rightarrow \square$

Conclusion:-

Therefore, Ravi receives a certification

4) Medical Attention Alert

Let: F = Ravi has a fever, M = Ravi needs medical attention

Logic:-

$$* F \rightarrow M$$

$$* \text{Goal: } M$$

CNF and Negation:-

$$* F \rightarrow M \equiv \neg F \vee M$$

$$* F \equiv F$$

$$\text{Negated Goal} = \neg M$$

Resolution:-

Resolve C_1 and C_2 : $(\neg F \vee M), F \rightarrow M$

Then!

$$M, \neg M \rightarrow \square$$

Conclusion:-

Ravi needs medical attention.

5) Vehicle Registration Process

Let:-

E = Car passed emissions test

R = Car is certified roadworthy

M = Car can be registered

Logic:-

* If car passes test, it is certified $\therefore E \rightarrow R$

* If car satisfies roadworthy it can be: $R \rightarrow G$

* Goal: G

Convert to CNF:-

$$E \rightarrow R = \neg E \vee R$$

$$R \rightarrow G = \neg R \vee G$$

Fact: E

Negated goal: $\neg G$

Resolution steps:-

Step 1:-

Resolve $C1$ and $C3$ $C1: \neg E \vee R$ $C3: E$

Step 2: Resolve R with $C2$

$$C2: \neg R \vee G$$

$\therefore G$

Step 3: Resolve G with $C4$

$$C4: \neg G$$

$\therefore \square$

Since the Empty clause ($\square$) is obtained, the negation of goal is impossible.

Therefore, the car can be registered.